Industrial Internship Report on

" Forecasting of Smart City Traffic Patterns"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Smart City Traffic Pattern Forecasting using Machine Learning." It focuses on analyzing traffic data from different junctions to predict future traffic flow. The project helps in identifying peak hours and supports better traffic management in smart cities. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface 3

2 Introduction 4

2.1 About UniConverge Technologies Pvt Ltd 4

2.2 About upskill Campus 7

2.3 Objective 9

2.4 Reference 9

2.5 Glossary 10

3 Problem Statement 11

4 Existing and Proposed solution 12

5 Proposed Design/ Model 13

5.1 High Level Diagram (if applicable) 13

5.2 Low Level Diagram (if applicable) ..... **Error! Bookmark not defined.**

5.3 Interfaces (if applicable) ..... **Error! Bookmark not defined.**

6 Performance Test ..... **Error! Bookmark not defined.**

6.1 Test Plan/ Test Cases ..... **Error! Bookmark not defined.**

6.2 Test Procedure ..... **Error! Bookmark not defined.**

6.3 Performance Outcome ..... **Error! Bookmark not defined.**

7 My learnings ..... **Error! Bookmark not defined.**

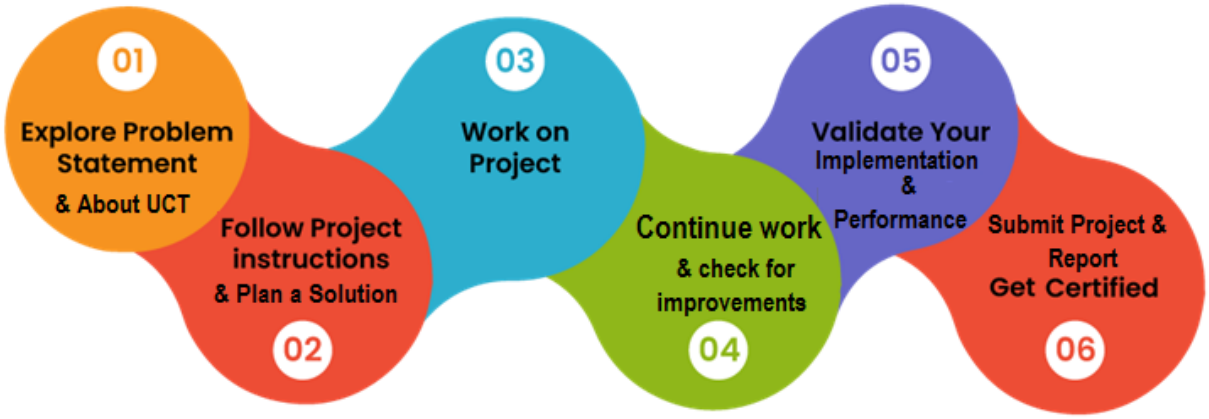
8 Future work scope ..... **Error! Bookmark not defined.**

1 Preface

This report presents the work completed during my internship in the domain of Data Science and Machine Learning. The internship provided an excellent opportunity to gain practical exposure and understand how theoretical concepts are applied to solve real-world problems.

The project undertaken during this internship is titled “Smart City Traffic Pattern Forecasting using Machine Learning.” The primary aim of this project is to analyze traffic data collected from different junctions and predict traffic flow using machine learning techniques.

With the increasing population and rapid urbanization, traffic congestion has become a major challenge in cities. This project focuses on developing a data-driven solution to analyze traffic patterns and support better traffic management and infrastructure planning.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

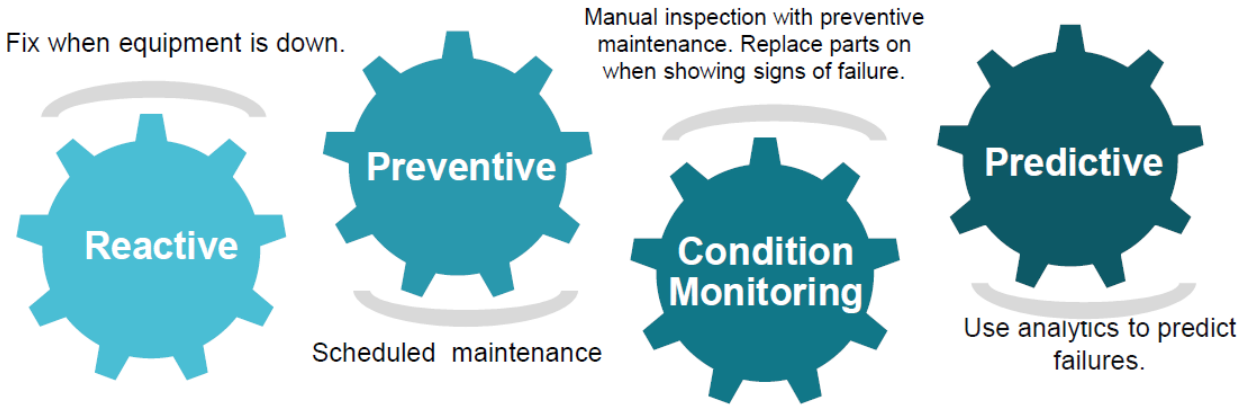
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

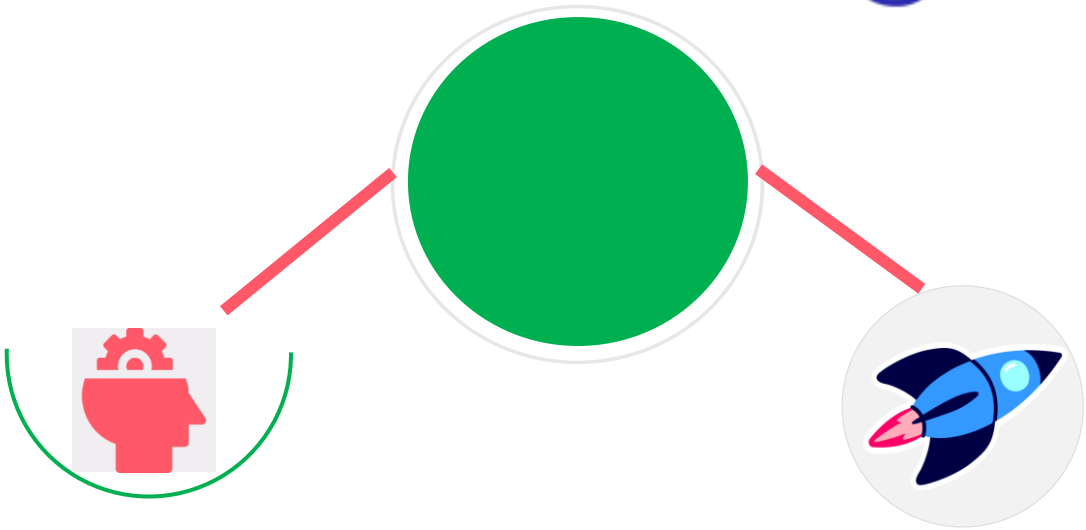
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

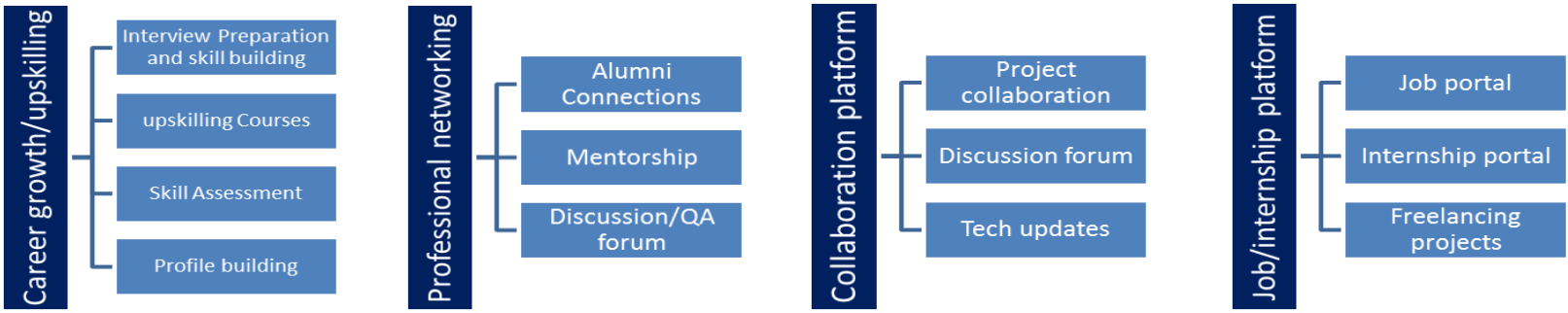
USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The primary objective of this internship project is to design and implement a predictive traffic forecasting solution for smart cities using machine learning. Specifically, we aim to collect and preprocess traffic sensor data, engineer relevant features, and train models to forecast short-term traffic patterns (e.g., vehicle flow, speed or volume) in urban road networks. By anticipating congestion and traffic flow ahead of time, the solution can help city planners optimize traffic management, reduce travel times, and lower pollution. The goal is to evaluate multiple approaches – from classical statistical methods to advanced spatio-temporal neural networks – and propose a high-performing, deployable forecasting model. Key project objectives include:

- Data Acquisition: Gather real or publicly available urban traffic datasets (e.g., METR-LA, PEMS-BAY) and/or IoT sensor feeds.
- Feature Engineering: Extract time-based (time-of-day, day-of-week, holidays) and spatial features (neighboring road sensors, network topology).
- Model Development: Explore statistical (e.g. ARIMA), classical ML (e.g. SVR, Random Forest), deep learning (LSTM, CNN), and graph neural network (GNN) models.
- Evaluation: Define metrics (RMSE, MAE, MAPE, R^2), perform train/validation splits (time-based cross-validation), and compare model performance.
- Deployment Considerations: Plan for real-time inference, scalability to large sensor networks, and compliance with data privacy/ethics.

2.5 Reference

This report is based on a combination of academic research, industry whitepapers, and official company sources. Primary references include peer-reviewed traffic forecasting studies, official UCT/upSkill documentation, and public traffic datasets (METR-LA, PEMS)[1][3]. Key insights and data (sensor counts, sample studies) are cited inline. References are listed at the end for full details

2.6 Glossary

Data Science: A field that involves analyzing data to extract useful insights	Data Science: A field that involves analyzing data to extract useful insights
Machine Learning: A technique where systems learn patterns from data	Machine Learning: A technique where systems learn patterns from data
Dataset: A structured collection of data used for analysis	Dataset: A structured collection of data used for analysis
Prediction: Estimating future outcomes based on historical data	Prediction: Estimating future outcomes based on historical data
Feature Engineering: Process of selecting and transforming variables for model building	Feature Engineering: Process of selecting and transforming variables for model building
Data Science: A field that involves analyzing data to extract useful insights	Data Science: A field that involves analyzing data to extract useful insights

3 Problem Statement

The government aims to transform cities into smart cities by improving infrastructure and public services. One of the major challenges faced is traffic congestion.

Traffic patterns vary depending on time, day, holidays, and special events. Managing traffic manually is inefficient and does not provide accurate predictions.

The objective of this project is to develop a machine learning model that can analyze traffic data from different junctions and forecast traffic patterns. This will help in improving traffic management, reducing congestion, and supporting better infrastructure planning.

4 Existing and Proposed solution

4.1 Existing Solution

- Traffic managed using fixed signal timings
- Manual monitoring of traffic flow
- No data-driven decision-making
- No prediction of traffic conditions
- Inefficient handling of peak traffic

4.2 Proposed Solution

- Use machine learning to analyze historical traffic data
- Extract time-based and location-based features
- Build predictive models for traffic forecasting
- Identify peak traffic hours and congestion patterns
- Support smart traffic management systems

4.3 Code submission (https://github.com/afshu05/-Internship_upskillCampus_Smart-City-Traffic-Forecasting)

4.4 Report submission (https://github.com/afshu05/-Internship_upskillCampus_Smart-City-Traffic-Forecasting)

5 Proposed Design/ Model

The proposed system uses a machine learning approach to predict traffic patterns.

5.1 Working Methodology

1. Data Collection

The dataset is obtained from Kaggle, containing traffic data from multiple junctions along with date and time information.

2. Data Preprocessing

- Convert DateTime column into proper format
- Handle missing or inconsistent data
- Clean dataset for analysis

3. Feature Engineering

New features are extracted from the DateTime column:

- Hour
- Day
- Month
- Day of the Week
- Junction

These features help the model understand traffic patterns.

4. Model Building

A Random Forest Regressor is used for prediction because it provides better accuracy and handles complex patterns.

5. Model Evaluation

The dataset is split into training and testing data. The model is evaluated using accuracy score.

6. Prediction

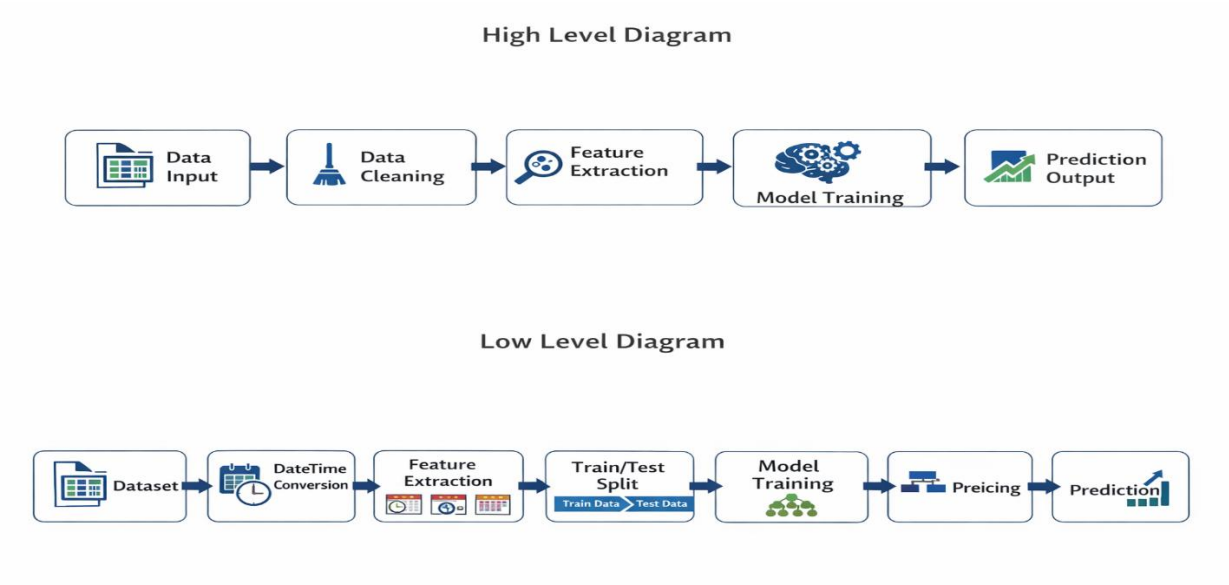
The trained model is used to predict traffic for future time intervals.

5.2 High Level Diagram

Input Data → Data Preprocessing → Feature Engineering → Model Training → Prediction Output

5.3 Low Level Diagram

Dataset → DateTime Conversion → Feature Extraction → Train/Test Split → Model Training → Prediction → Output



5.4 Interfaces

- Programming Language: Python
- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
- Input: CSV Dataset
- Output: Predicted Traffic Values

6. Performance Test

6.1 Test Plan / Test Cases

- Train model using training dataset
- Test model using unseen data
- Compare predicted values with actual values

6.2 Test Procedure

- Split dataset into training and testing sets
- Train the model using Random Forest algorithm
- Evaluate model performance using accuracy score

6.3 Performance Outcome

- The model successfully predicted traffic patterns
- The predictions were close to actual values
- The model performed well for basic implementation

7. My Learnings

During this internship, I gained valuable knowledge and skills:

- Learned Data Science and Machine Learning concepts
- Understood data preprocessing and feature engineering
- Gained experience in working with real-world datasets
- Learned how to build and evaluate machine learning models
- Improved problem-solving and analytical skills

8. Future Work Scope

The project can be further improved in the following ways:

- Use real-time traffic data
- Apply advanced models such as Deep Learning
- Improve model accuracy
- Develop a web-based dashboard
- Integrate with smart city systems

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